



COLLEGE OF INFORMATION SCIENCES AND COMPUTING

Bachelor of Science in Information Technology
ITCC47 – Data Structures and Algorithms
1st Semester, 2026-2027

VISION A leading ASEAN University actively committed to the total development of people for a globally sustainable environment and humane society.	MISSION To advance the frontier of knowledge through internationalization of education and equitable access to quality instruction, research, extension, and production for economic prosperity, moral integrity, social and cultural sensitivity, and environmental consciousness through equitable access to quality instruction, research, extension, and production.	QUALITY POLICY Working towards total customer satisfaction for nation building and global competitiveness, Central Mindanao University commits to provide quality services in instruction, research, extension and production through continual improvement of quality management system processes.
PROGRAM INFORMATION Name of Program : Bachelor of Science in Information Technology CHED CMO Reference : CMO no. 25, s. 2015 BOR Approval : No. 10 s. 2018		
PROGRAM OUTCOMES		
Common to All Programs in All Types of Schools		
a	Articulate and discuss the latest developments in the specific field of practice.	
b	Effectively communicate orally and in writing using both English and Filipino.	
c	Work effectively and independently in multidisciplinary and multi-cultural teams.	
d	Act in recognition of professional, social, and ethical responsibility.	
e	Preserve and promote “Filipino historical and cultural heritage”	
Common to the Discipline		
f	Analyse complex problems, and identify and define the computing requirements needed to design an appropriate solution.	
g	Apply computing and other knowledge domains to address real-world problems.	
h	Design and develop computing solutions using a system system-level perspective.	
i	Utilize modern computing tools	
Specific to Bachelor of Science in Information Technology		
j	Apply Knowledge of computing, science and mathematics appropriate to the discipline	
k	Understand best practices and standards and their applications	



l	Analyse complex problems, and identify and define the computing requirements appropriate to its solution
m	Identify and analyse user needs and take them into account in the selection, creation, evaluation and administration of computer-based systems
n	Design, implement, and evaluate computer-based systems, processes, components, or programs to meet desired needs and requirements under various constraints
o	Integrate IT-based solutions into the user environment friendly
p	Apply Knowledge through the use of current techniques, skills, tools and practices necessary for the IT profession
q	Function effectively as a leader or member of development team recognizing the different tools of a team need to accomplish a common goal.
r	Assist in the creation of an effective IT project plan.
s	Communicate effectively with the computing community and society at large about complex computing activities through logical writing, presentations and clear instructions
t	Analyse the global and local impact of computing information technologies on individuals, organizations and society
u	Understand professional, ethical, legal, security and social issues in responsibilities in the utilization of information technology
v	Recognize the need for and engage in planning, self-learning and improving performance as a foundation for continuing professional development.

COURSE INFORMATION					
Course Name	Data Structures and Algorithms			Course Code	ITCC47
Pre-requisite Subject	ITCC 44	Course Credit	3.0 units	Pre-requisite Subject	ITCC44
Course Description: Data Structures and Algorithms focuses on the design, implementation, and analysis of efficient program solutions through appropriate data organization and algorithmic strategies. It introduces fundamental data structures such as arrays, linked lists, stacks, queues, trees, and graphs, and applies them to practical problems involving searching, sorting, traversal, shortest paths, greedy methods, and basic dynamic programming. The course emphasizes computational thinking, efficiency analysis, implementation correctness, and the use of structured laboratory testing to solve real-world IT problems.					



Course Learning Outcomes (CLO)	Program Outcomes Code																						
	a	b	c	d	e	f	g	h	i	j	k	L	m	n	o	p	q	r	s	t	u	v	
Program Outcomes (PO) addressed by the course	D	D	D			D	D	D	D	D	D	D	D	D	D	D	D	D	D	D		D	
Program Outcomes addressed by the Course Learning Outcomes:	Program Outcomes Code																						
	a	b	c	d	e	f	g	h	i	j	k	L	m	n	o	p	q	r	s	t	u	v	
CLO 1. Explain the concept of data organization on applications and systems in general.	E		E			E	E	E	E	E	E	E	E	E	E	E	E	E	E		E		
CLO 2. Analyze algorithms and quantify their efficiency in handling an arbitrary set of inputs and processing an arbitrary amount of output.	D	D	D			D	D	D	D	D	D	D	D	D	D	D	D	D	D		D		
CLO 3. Write down pseudocodes and interpret its content into a personal choice of a computer language.	D		D			D	D	D	D	D	D	D	D	D	D	D	D	D	D		D		
CLO 4. Implement data structures and algorithms that will sufficiently solve programming problems.	D		D			D	D	D	D	D	D	D	D	D	D	D	D	D	D		D		
CLO 5. Improve algorithm efficiency when creating program solutions.	D		D			D	D	D	D	D	D	D	D	D	D	D	D	D	D		D		
CLO 6. Implement and modify the behavior of a data structure according to the objects/items to be stored.	D		D			D	D	D	D	D	D	D	D	D	D	D	D	D	D		D		

***Level:** I – Introduced to concepts/principles E - Enabling D – Demonstrated with minimal supervision (*Please refer to Program CMO*)



LEARNING PLAN							
Time Allotment (Week)	Course Learning Outcomes (CLOs)	Intended Learning Outcomes (ILOs)	Learning Content/Topics	Teaching and Learning Activities		Assessment Tasks	References
				Teaching Activities	Learning Activities		
0.5 Weeks		1. Familiarized with CMU mission and vision, course context, grading system, and classroom policies. 2. Explained the goals of the college and objectives of the department; and 3. Acquainted with each other and level off expectations. 4. Familiarized with the rules and regulations in the laboratory classes.	• Course Orientation • Course Syllabus • Class Policies	• Sharing of Online Reading Materials through LMS	• Reading Assignment		• Course Syllabus • CMU Code • Operation Manual for ICA Computer Lab.
1.5 Weeks	CLO 1 CLO 2 CLO 3	1. Explain the role of algorithms in solving computing problems 2. Create pseudocode for simple computational tasks 3. Translate pseudocode into a working program 4. Trace an algorithm manually using sample input 5. Identify simple time complexity patterns such as $O(1)$, $O(n)$, and $O(n^2)$ 6. Follow exact input/output specifications required for automated testing	MODULE 1: Algorithmic Thinking, Pseudocode, Exact I/O, Complexity Basics A. Algorithmic Thinking B. Pseudocode and Flow of Logic C. Exact Input and Output Discipline D. Complexity Basics	• Face-to-Face lecture • Teacher Demonstration • Sharing of Online Reading Materials through LMS	• Discussion • Pair Programming • Reading Assignment	• Chapter Quiz • Laboratory Activity	1,2,3



2 Weeks	CLO 1 CLO 2 CLO 3 CLO 4 CLO 5	1. Use arrays or lists to store and process multiple values 2. Implement traversal, search, insertion, and deletion operations on list-based data 3. Compare linear search and binary search based on input conditions 4. Apply simple sorting algorithms to organize data 5. Analyze the efficiency of array/list operations 6. Solve frequency-counting and duplicate-detection problems	MODULE 2: Array, Lists, Searching, Sorting Foundations A. Arrays and Lists B. Searching Algorithms C. Sorting Foundations D. Frequency and Counting Problems	• Face-to-Face lecture • Teacher Demonstration • Sharing of Online Reading Materials through LMS	• Pair Programming • Reading Assignment	• Chapter Quiz • Laboratory Activity	1,2,3
2 Weeks	CLO 1 CLO 2 CLO 3 CLO 4 CLO 6	1. Explain how node-based data organization differs from any array-based storage 2. Implement a singly linked list 3. Perform basic linked list operations such as insert, search, update, delete, and display 4. Store and manipulate object-based records in a linked list 5. Compare the practical advantages and limitations of arrays and linked lists 6. Modify linked list behavior according to the requirements of a given problem	MODULE 3: Linked Lists and Dynamic Data Organization A. Node-based Storage B. Singly Linked Lists C. Object-based Linked Lists D. Array List vs Linked List	• Face-to-Face lecture • Teacher Demonstration • Sharing of Online Reading Materials through LMS	• Pair Programming • Reading Assignment	• Chapter Quiz • Project 0 – ASEAN Phonebook	1,2,3
2 Weeks	CLO 2 CLO 3 CLO 4 CLO 5 CLO 6	1. Differentiate stacks, queues, and dequeues based on access behavior 2. Implement stack, queue, and deque operations using arrays or linked structures 3. Apply stacks to validation and reversal problems 4. Apply queues for scheduling and ordered-processing problems 5. Select the appropriate linear data structure for a given scenario 6. Analyze the efficiency of stack and queue operations	MODULE 4: Stacks, Queues, and Deques A. Stacks B. Queues C. Deques D. Application Problems	• Face-to-Face lecture • Teacher Demonstration • Sharing of Online Reading Materials through LMS	• Pair Programming • Reading Assignment	• Chapter Quiz • Project 1 – Listen to the Music	1,2,3
MIDTERM WEEK							



2 Weeks	CLO 4 CLO 5 CLO 6	1. Explain recursion using base cases and recursive cases 2. Trace recursive function calls 3. Implement simple recursive algorithms 4. Apply divide-and-conquer thinking to search and sorting problems 5. Implement a binary search on sorted data 6. Compare recursive and iterative solutions based on readability and efficiency	MODULE 5: Recursion, Divide-and-Conquer, and Binary Search A. Recursion B. Divide-and-Conquer C. Binary Search D. Sorting Connection	• Face-to-Face lecture • Teacher demonstration • Sharing of Reading Materials through LMS	• Guided group discussion/lecture • Pair Programming • Reading Activities	• Chapter Quiz • Laboratory Activity	1,2,3
2Weeks	CLO 1 CLO 2 CLO 4 CLO 5 CLO 6	1. Explain tree terminology and common tree applications 2. Implement a binary tree or binary search tree 3. Perform preorder, in-order, post-order, and level-order traversals 4. Apply BST insertion and search operations 5. Explain how tree height affects search efficiency 6. Recognize why balancing is useful in maintaining efficient tree operations	MODULE 6: Trees and Binary Search Trees A. Tree Concepts B. Binary Trees C. Tree Traversals D. Binary Search Trees E. Balanced Trees as a Concept	• Face-to-Face lecture • Teacher demonstration • Sharing of Reading Materials through LMS	• Guided group discussion/lecture • Pair Programming • Reading Activities	• Chapter Quiz • Laboratory Activity	1,2,3
2 Weeks	CLO 1 CLO 2 CLO 4 CLO 5 CLO 6	1. Explain graphs using vertices, edges, and relationships 2. Represent graphs using adjacency matrices and adjacency lists 3. Implement BFS and DFS traversal 4. Use BFS to solve shortest-path problems in unweighted graphs 5. Apply graph traversal to real-world IT and simulation problems 6. Analyze graph traversal efficiency in terms of vertices and edges	MODULE 7: Graphs, BFS, DFS, and Shortest Path Problems A. Graph Concepts B. Graph Representation C. Graph Traversal D. Shortest Path E. Applied Problems	• Face-to-Face lecture • Teacher demonstration • Sharing of Reading Materials through LMS	• Pair Programming • Reading Activities	• Chapter Quiz • Project 2 – Pathfinder	1,2,3



2 Weeks	CLO 2 CLO 3 CLO 4 CLO 5	1. Explain the greedy approach to algorithm design 2. Apply greedy logic to simple optimization problems 3. Identify cases where greedy solutions fail 4. Explain the basic idea of dynamic programming 5. Use table-based or memorized solutions for simple repeated-subproblem tasks 6. Compare greedy and dynamic programming solutions for coin-change problems	MODULE 8: Greedy Algorithms and Basic Dynamic Programming A. Greedy Algorithms B. Greedy Applications C. Dynamic Programming Basics D. DP Applications	• Face-to-Face lecture • Teacher demonstration • Sharing of Reading Materials through LMS	• Pair Programming • Reading Activities	• Chapter Quiz • Project 3 – The Greedy Merchant	1,2,3
FINAL TERM WEEK							

Total Contact Hours: 18 weeks/32 Hours

No.	RESOURCE MATERIALS		
1	Computer/Laptop	3	Books
2	Projector	4	Software: Visual Studio Code, PyCharm
No.	REFERENCES AND SUPPLEMENTARY READINGS		
1	Horowitz, E., Sahni, S., Rajasekaran, S. (1998). <i>Computer Algorithms</i> . Computer Science Press.		
2	Preiss, B. (2003). <i>Data Structures and Algorithms with Object-Oriented Design Patterns in Python</i> .		
3	Hug, J. (2021). <i>CS 61B Data Structures</i> . Accessed from: https://sp21.datastructur.es/		

COURSE REQUIREMENTS AND CLASS POLICY

- 1) No usage of mobile cellphones during classes.
- 2) Students are not allowed to enter lecture class after 15 minutes or 30 minutes depending on block schedule.
- 3) Students are not allowed to enter the laboratory class after 45 minutes of their block schedule.

Evaluation Guide and Grading System

COs	Assessment Tasks (Please specify type of assessment)	Weight in Percent/Type of Assessment	Minimum Average for Satisfactory Rating	Target and Standards (Refer to College Target & Standards)
Lecture				
CLO 1-6	Chapter Quizzes	50%	50%	70% of students got at least 50% rating
	Term Exam	50%		



	Assessment 1 Grade	100%		
CLO 4-6	Chapter Quizzes	50%	50%	0
	Term Exam	50%		
	Assessment 2 Grade	100%		
Laboratory				
	Laboratory Activities	100%	50%	70% of students got at least 50% rating
	Assessment 1 Grade	100%		
	Laboratory Activities	20%	50%	70% of students got at least 50% rating
	Laboratory Project	40%		
	Practical Exam	40%		
	Assessment 2 Grade	100%		

E.g., for 1 unit Lec., 2 units Lab. Course

Midterm Grade: 2/3 Assessment 1 Grade in Lec, 1/3 Assessment 1 Grade in Lab

Final term Grade: 2/3 Assessment 2 Grade in Lec, 1/3 Assessment 2 Grade in Lab

Final Grade: 1/3 Midterm Grade, 2/3 Final term Grade

The final grades will correspond to the weighted average scores shown below:

Range	94-100	88-93	82-87	76-81	70-75	64-69	58-63	52-57	50-51	30-49	29-below	Incomplete
Grade	1.0	1.25	1.50	1.75	2.0	2.25	2.50	2.75	3.0	4.0 (removal)	5.0	INC

Revision History

Revision number	Date of Revision	Date of implementation	Highlights of Revision (Eg. Revision of grading system, policy, etc.)
1	August 7, 2023	August 11, 2023	Initial implementation.
2	July 22, 2024	August 5, 2024	Revision of course content, course learning outcomes, grading system, and references.
3	July 27, 2026	August 3, 2026	Revision of course content, course learning outcomes, course policy, and references.

Preparation and Review

	Name	Signature	Date Signed
Prepared by	NEIL JOHN J. JOMAYA		
	Faculty in-Charge, Lecture		
	RIZZA MAE G. DAGOC		
	Faculty in-Charge, Laboratory		



Reviewed by	MINDELL REY C. AGUIRRE		
	Chairperson, IT Department		

Approval

Approved by:	JOHN D. TAJONES		
	College Dean		

- If the syllabus is prepared by the department chair, it should be reviewed by a senior faculty member of the department. If the syllabus is prepared by the dean, it should be reviewed by the department chair.